

# **SIMATS ENGINEERING**

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**

## **CO3-AT1 CASE STUDY REPORT**

**Submitted for**

**DSA04 – Fundamentals of Data Science**

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# Statistics Case Study Report

## *Descriptive Statistics, Variability & Z-Scores*

Case Studies 6 – 10 | CO3 AT01

### CASE STUDY 6 — DETECTING AN UNUSUAL CUSTOMER

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400. A particular customer spends ₹3,200.

| Given Data                      | Value  |
|---------------------------------|--------|
| Mean ( $\mu$ )                  | ₹2,000 |
| Standard Deviation ( $\sigma$ ) | ₹400   |
| Customer's Spending ( $x$ )     | ₹3,200 |

**Q1. Calculate the customer's z-score.**

Formula:  $z = (x - \mu) / \sigma$

$$z = (3200 - 2000) / 400$$

$$z = 1200 / 400$$

► Answer:  $z = 3.0$

**Q2. Is the customer's spending unusually high compared with the average?**

► Answer: Yes. A z-score of 3.0 means the customer's spending is 3 standard deviations above the mean. Under the empirical (68-95-99.7) rule, values beyond  $\pm 3\sigma$  occur less than about 0.3% of the time, so this spending is statistically very unusual (an extreme high outlier).

**Q3. What would a negative z-score mean in this context?**

► Answer: A negative z-score would mean the customer spent less than the average (₹2,000) — i.e., their spending falls below the mean. The more negative the z-score, the further below average the spending is.

**Q4. If another customer has a z-score of  $-1.5$ , what does it tell you?**

$$x = \mu + z\sigma = 2000 + (-1.5 \times 400) = 2000 - 600 = ₹1,400$$

► Answer: This customer spent ₹1,400, which is 1.5 standard deviations below the mean. This is lower than typical spending, but since it lies within  $\pm 2\sigma$ , it is NOT considered statistically unusual — just a below-average spender.

## CASE STUDY 7 — DELIVERY TIME ANALYSIS

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A delivery company records delivery times (in minutes) over 10 deliveries:

25, 27, 28, 29, 30, 30, 31, 32, 34, 60 (n = 10, already sorted)

### Q1. Calculate the mean and median.

$$\text{Sum} = 25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$$

$$\text{Mean} = 326 / 10 = 32.6 \text{ minutes}$$

$$\text{Median} = \text{average of 5th \& 6th values} = (30 + 30) / 2 = 30 \text{ minutes}$$

► Answer: Mean = 32.6 min, Median = 30 min

### Q2. Find Q1 and Q3.

Lower half (1st–5th): 25, 27, 28, 29, 30 → Q1 = median = 28

Upper half (6th–10th): 30, 31, 32, 34, 60 → Q3 = median = 32

► Answer: Q1 = 28 minutes, Q3 = 32 minutes

### Q3. Calculate the IQR.

$$\text{IQR} = Q3 - Q1 = 32 - 28 = 4$$

► Answer: IQR = 4 minutes

### Q4. Check for outliers using the $1.5 \times \text{IQR}$ rule.

$$\text{Lower bound} = Q1 - 1.5 \times \text{IQR} = 28 - 6 = 22$$

$$\text{Upper bound} = Q3 + 1.5 \times \text{IQR} = 32 + 6 = 38$$

► Answer: Any value below 22 or above 38 is an outlier.

The value 60 exceeds 38, so 60 IS an outlier.

### Q5. Explain whether the mean or median gives a better description of the typical delivery time.

► Answer: The median (30 minutes) is the better representative of typical delivery time. The mean (32.6) is inflated by the extreme outlier (60), which pulls it upward. Since the median is resistant to outliers, it better reflects what a 'typical' delivery actually looks like.

## CASE STUDY 8 — TWO CITIES' TEMPERATURES

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Average daily temperatures (°C) over 7 days:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

**Q1. Calculate the mean temperature for each city.**

City A:  $\text{Sum} = 28+29+30+30+31+30+32 = 210 \rightarrow \text{Mean} = 210/7 = 30\text{ }^{\circ}\text{C}$

City B:  $\text{Sum} = 20+25+30+35+30+25+45 = 210 \rightarrow \text{Mean} = 210/7 = 30\text{ }^{\circ}\text{C}$

► **Answer: Mean(A) = 30 °C, Mean(B) = 30 °C (identical!)**

**Q2. Calculate the variance for each city.**

City A deviations from mean(30): -2,-1,0,0,1,0,2

Squared: 4,1,0,0,1,0,4  $\rightarrow \text{Sum} = 10$

$\text{Var(A)} = 10 / 7 = 1.43\text{ }^{\circ}\text{C}^2$

City B deviations from mean(30): -10,-5,0,5,0,-5,15

Squared: 100,25,0,25,0,25,225  $\rightarrow \text{Sum} = 400$

$\text{Var(B)} = 400 / 7 = 57.14\text{ }^{\circ}\text{C}^2$

► **Answer: Var(A)  $\approx 1.43\text{ }^{\circ}\text{C}^2 \rightarrow \text{SD(A)} \approx 1.20\text{ }^{\circ}\text{C}$**

► **Var(B)  $\approx 57.14\text{ }^{\circ}\text{C}^2 \rightarrow \text{SD(B)} \approx 7.56\text{ }^{\circ}\text{C}$**

**Q3. Which city has greater temperature variability?**

► **Answer: City B has much greater variability (Var  $\approx 57.14$  vs 1.43).**

**Q4. Why can two datasets have the same mean but different standard deviations?**

► **Answer: The mean only measures the central location (average value) of a dataset — it says nothing about how spread out the values are around that centre. Two datasets can be centred at the same point while one clusters tightly around it (low SD) and the other swings widely above and below it (high SD). Standard deviation captures this spread, which mean alone cannot.**

**Q5. Which city has more predictable temperatures?**

► **Answer: City A is more predictable — its low standard deviation ( $\approx 1.20\text{ }^{\circ}\text{C}$ ) means daily temperatures stay close to the average, unlike City B's volatile swings ( $\approx 7.56\text{ }^{\circ}\text{C}$ ).**

## CASE STUDY 9 — DATA CENTER CPU USAGE

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A data scientist records CPU usage percentages over 10 readings:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90 (n = 10, already sorted)

### Q1. Calculate mean, median, and mode.

$$\text{Sum} = 40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513$$

$$\text{Mean} = 513 / 10 = 51.3$$

$$\text{Median} = \text{average of 5th \& 6th values} = (46 + 48) / 2 = 47$$

$$\text{Mode} = 45 \text{ (appears twice — the only repeated value)}$$

► Answer: Mean = 51.3%, Median = 47%, Mode = 45%

### Q2. Calculate the range.

$$\text{Range} = \text{Max} - \text{Min} = 90 - 40 = 50$$

► Answer: Range = 50%

### Q3. Find Q1, Q3 and IQR.

$$\text{Lower half (1st–5th): } 40, 42, 45, 45, 46 \rightarrow Q1 = \text{median} = 45$$

$$\text{Upper half (6th–10th): } 48, 50, 52, 55, 90 \rightarrow Q3 = \text{median} = 52$$

$$\text{IQR} = Q3 - Q1 = 52 - 45 = 7$$

► Answer: Q1 = 45, Q3 = 52, IQR = 7

### Q4. Identify possible outliers.

$$\text{Lower bound} = Q1 - 1.5 \times \text{IQR} = 45 - 10.5 = 34.5$$

$$\text{Upper bound} = Q3 + 1.5 \times \text{IQR} = 52 + 10.5 = 62.5$$

► Answer: Values outside [34.5, 62.5] are outliers.

90 exceeds 62.5, so 90 IS an outlier.

### Q5. Determine the direction of skewness.

► Answer: Mean (51.3) > Median (47), so the data is right-skewed (positively skewed). The outlier (90) pulls the mean toward the higher tail, while the median stays closer to the bulk of the data.

## CASE STUDY 10 — COMPARING TWO DATASETS

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Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

**Q1. Calculate the mean of both datasets.**

Dataset A:  $(10+20+30+40+50)/5 = 150/5 = 30$

Dataset B:  $(28+29+30+31+32)/5 = 150/5 = 30$

► **Answer: Mean(A) = 30, Mean(B) = 30 (identical!)**

**Q2. Calculate the variance of both.**

Dataset A deviations: -20,-10,0,10,20 → squared: 400,100,0,100,400 = 1000

Var(A) =  $1000 / 5 = 200$

Dataset B deviations: -2,-1,0,1,2 → squared: 4,1,0,1,4 = 10

Var(B) =  $10 / 5 = 2$

► **Answer: Var(A) = 200, Var(B) = 2**

**Q3. Calculate the standard deviation of both.**

SD(A) =  $\sqrt{200} \approx 14.14$

SD(B) =  $\sqrt{2} \approx 1.41$

► **Answer: SD(A)  $\approx$  14.14, SD(B)  $\approx$  1.41**

**Q4. Which dataset has greater spread?**

► **Answer: Dataset A has far greater spread (SD  $\approx$ 14.14 vs  $\approx$ 1.41; Var 200 vs 2).**

**Q5. If both datasets represent model errors, which model appears more consistent?**

► **Answer: Dataset B represents the more consistent model. Its errors cluster tightly around 30 (low variance/SD), meaning predictions are stable and reliable, whereas Dataset A's errors vary widely, indicating unpredictable performance even though both average the same error.**

**Q6. Explain why mean alone is not enough to compare datasets.**

► **Answer: Both datasets share an identical mean (30), yet they behave very differently. Dataset A is widely spread (10 to 50) while Dataset B is tightly clustered (28 to 32). The mean summarizes only the 'centre' of the data — it hides variability, consistency, and risk. Measures of spread (variance, standard deviation, IQR, range) are essential alongside the mean to fully understand and compare datasets.**

